

# CCNA Practice Project

Multi-Department Network with VLSM Subnetting,  
DHCP Services, and Static Routing

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|                 |                                                                         |
|-----------------|-------------------------------------------------------------------------|
| Simulation Tool | Cisco Packet Tracer                                                     |
| Devices Used    | 3 x Router 2621XM, 3 x Switch 2960-24TT, 6 End Devices                  |
| Departments     | HR (50 PCs), Production (120 PCs), Sales (250 PCs)                      |
| Key Concepts    | VLSM, DHCP Pools, Default Gateway, Static Routing, Connectivity Testing |

# 1. Project Objective

This project simulates a small company network divided into three departments — **HR**, **Production**, and **Sales** — each connected through its own router and switch. The goal is to design an IP addressing plan using **Variable Length Subnet Masking (VLSM)** so that each department gets a subnet sized to its actual number of hosts, configure each router to hand out addresses automatically using **DHCP**, connect the three routers together with **static routes**, and finally verify that every device can reach every other device across the network using ping and ipconfig.

## 2. Network Topology

The network consists of three routers (Router0, Router1, Router2), each connected to its own access-layer switch and a small LAN. The routers are chained together in series (Router0 ↔ Router1 ↔ Router2) using point-to-point FastEthernet links, so traffic between HR and Sales must pass through Router1 in the middle.

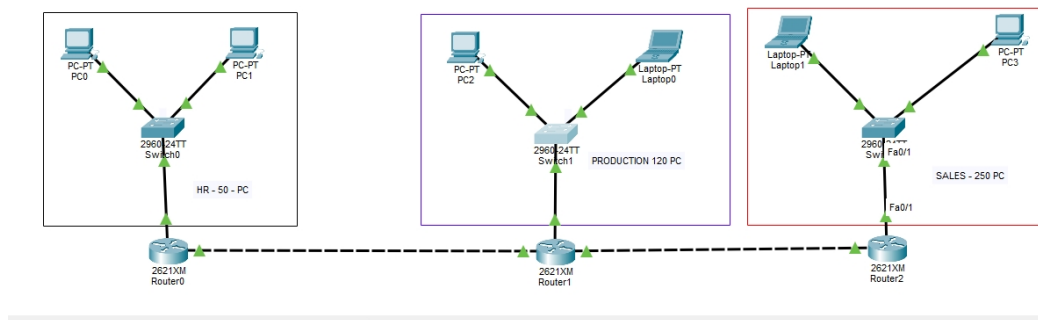


Figure 1 — Overall topology: HR (Router0/Switch0), Production (Router1/Switch1), and Sales (Router2/Switch2), interconnected in a chain.

## 3. IP Addressing Plan (VLSM)

Instead of giving every department the same size subnet, VLSM was used to match subnet size to the number of hosts needed, which avoids wasting IP addresses on the smaller departments.

| Department           | Hosts Needed | Network / Mask                      | Usable Range            | Default Gateway |
|----------------------|--------------|-------------------------------------|-------------------------|-----------------|
| Sales                | 250 PCs      | 50.0.0.0 /24<br>(255.255.255.0)     | 50.0.0.1 - 50.0.0.254   | 50.0.0.1        |
| Production           | 120 PCs      | 50.0.1.0 /25<br>(255.255.255.128)   | 50.0.1.1 - 50.0.1.126   | 50.0.1.1        |
| HR                   | 50 PCs       | 50.0.1.128 /26<br>(255.255.255.192) | 50.0.1.129 - 50.0.1.190 | 50.0.1.129      |
| Router0-Router1 link | 2 (WAN)      | 50.0.1.192 /30                      | 50.0.1.193 - 50.0.1.194 | -               |
| Router1-Router2 link | 2 (WAN)      | 50.0.1.196 /30                      | 50.0.1.197 - 50.0.1.198 | -               |

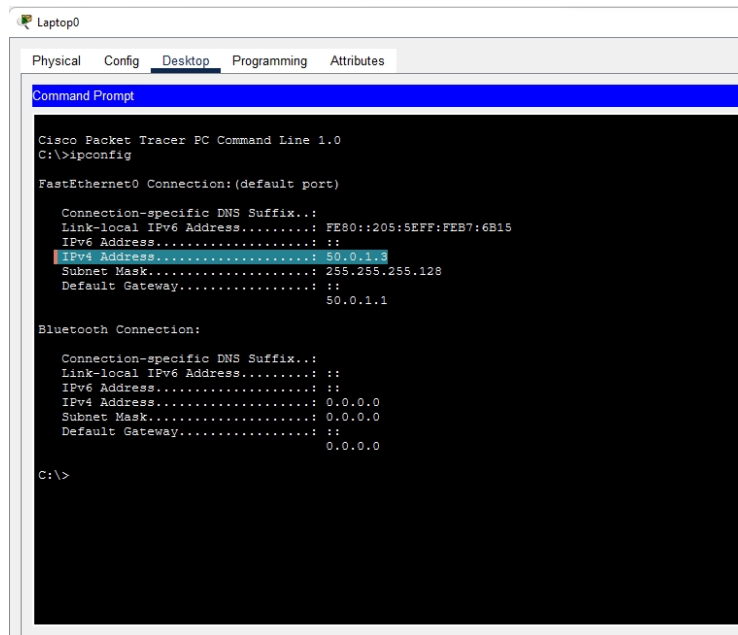
Note the largest department (Sales, 250 hosts) was given the biggest block first (a /24), then Production (120 hosts) took a /25 carved out of the remaining space, and HR (50 hosts) took a /26. The two router-to-router

links only need 2 usable addresses each, so they were assigned the smallest possible subnets, /30 (point-to-point links).

## 4. End-Device IP Configuration (ipconfig)

Each PC and laptop was set to obtain its address automatically via DHCP from its local router. Running ipconfig on every end device confirms it received an address inside the correct department subnet, with the correct subnet mask and default gateway.

### 4.1 Production Department



```
Laptop0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

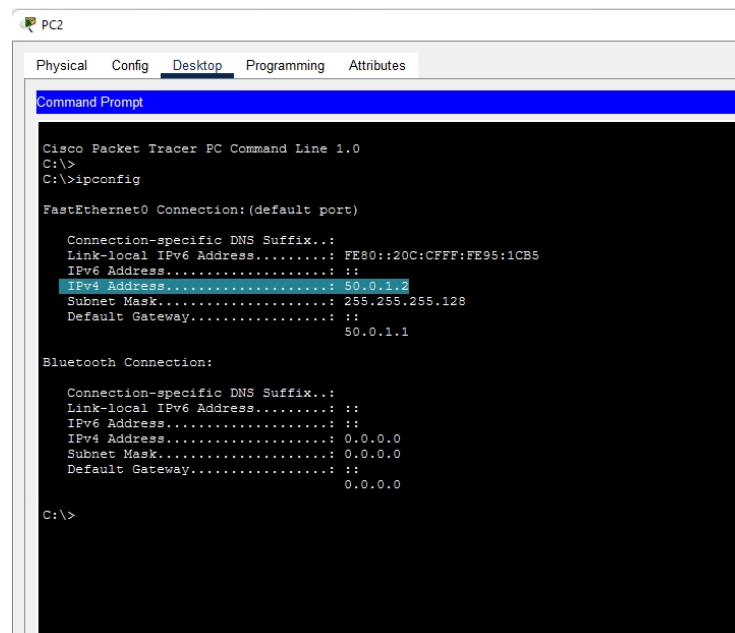
    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::205:5EFF:FEB7:6B15
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 50.0.1.3
    Subnet Mask . . . . .: 255.255.255.128
    Default Gateway . . . . .: 50.0.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

C:\>
```

Figure 2 — Laptop0 (Production) received 50.0.1.3 /25, gateway 50.0.1.1.



```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20C:CFFF:FE95:1CB5
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 50.0.1.2
    Subnet Mask . . . . .: 255.255.255.128
    Default Gateway . . . . .: 50.0.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

C:\>
```

Figure 3 — PC2 (Production) received 50.0.1.2 /25, gateway 50.0.1.1.

### 4.2 HR Department

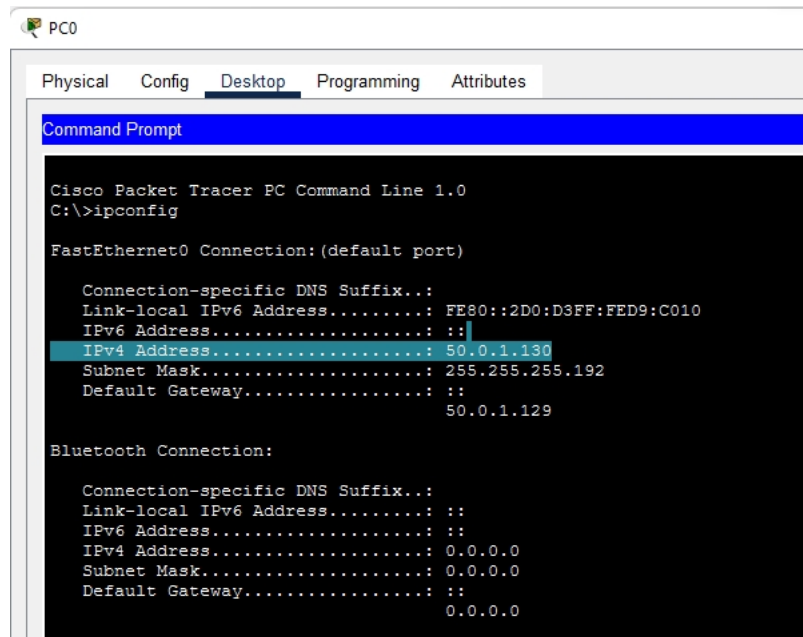


Figure 4 — PC0 (HR) received 50.0.1.130 /26, gateway 50.0.1.129.

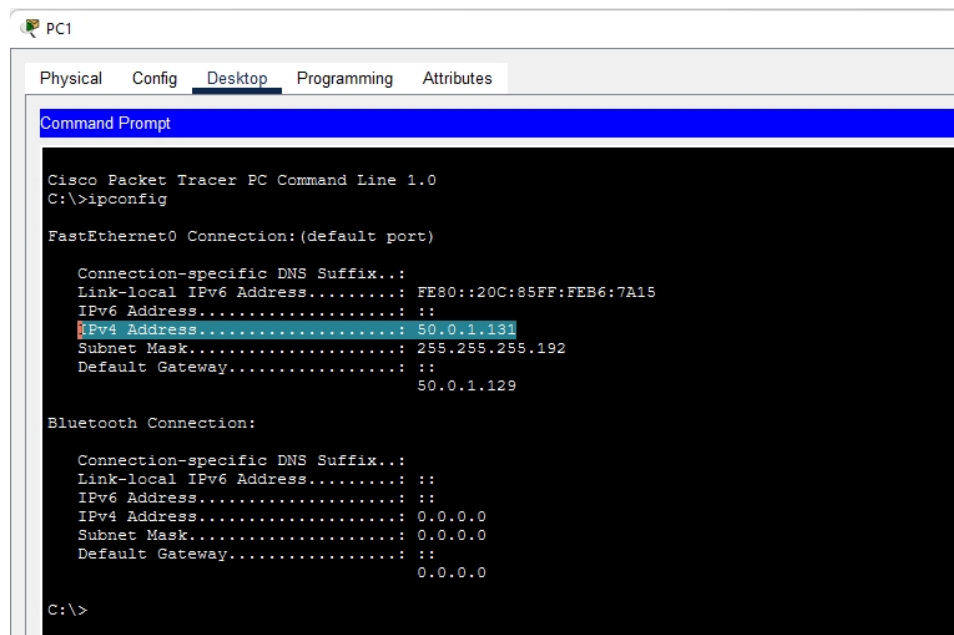
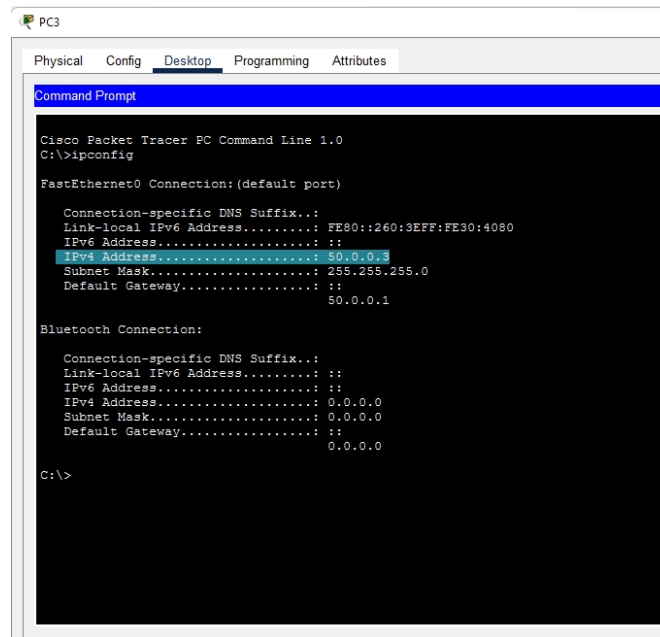


Figure 5 — PC1 (HR) received 50.0.1.131 /26, gateway 50.0.1.129.

## 4.3 Sales Department



The screenshot shows the Command Prompt window for PC3 in Cisco Packet Tracer. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt displays the output of the 'ipconfig' command. It shows the configuration for the FastEthernet0 interface, including the IPv4 address 50.0.0.3, subnet mask 255.255.255.0, and default gateway 50.0.0.1. The Bluetooth connection is also shown with its respective configuration.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

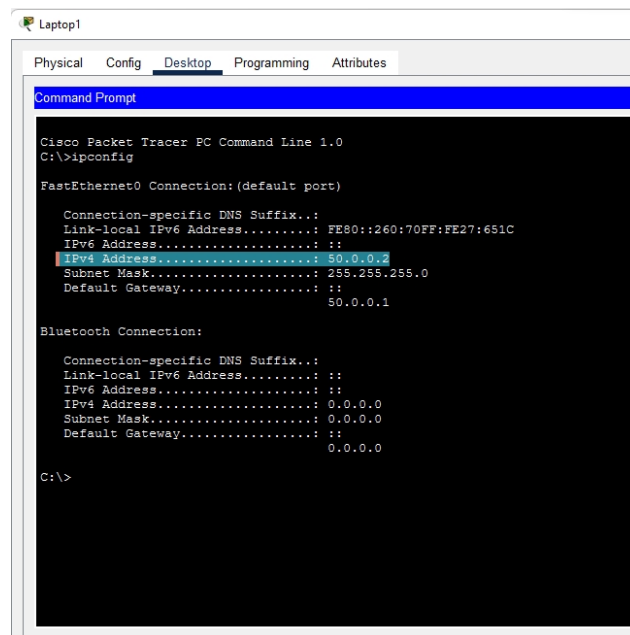
    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::260:3EFF:FE30:4080
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 50.0.0.3
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   50.0.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```

Figure 6 — PC3 (Sales) received 50.0.0.3 /24, gateway 50.0.0.1.



The screenshot shows the Command Prompt window for Laptop1 in Cisco Packet Tracer. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt displays the output of the 'ipconfig' command. It shows the configuration for the FastEthernet0 interface, including the IPv4 address 50.0.0.2, subnet mask 255.255.255.0, and default gateway 50.0.0.1. The Bluetooth connection is also shown with its respective configuration.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::260:70FF:FE27:651C
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 50.0.0.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   50.0.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

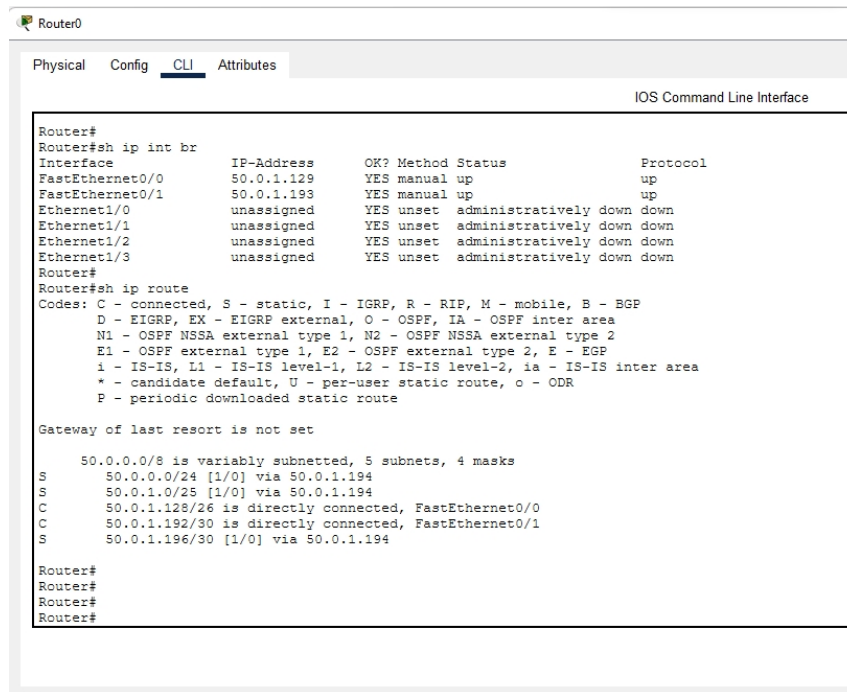
C:\>
```

Figure 7 — Laptop1 (Sales) received 50.0.0.2 /24, gateway 50.0.0.1.

## 5. Router Configuration Verification

### 5.1 Router0 (HR Router)

Router0 connects the HR LAN (FastEthernet0/0, 50.0.1.129) to Router1 across the 50.0.1.192/30 link (FastEthernet0/1, 50.0.1.193). Since no dynamic routing protocol is running, static routes point every remote subnet (Production, Sales, and the far WAN link) toward 50.0.1.194, the next hop on Router1.



The screenshot shows the Router0 CLI interface with the 'CLI' tab selected. The output of the 'sh ip int br' command shows the status of interfaces FastEthernet0/0, FastEthernet0/1, and Ethernet1/0 through Ethernet1/3. FastEthernet0/0 and FastEthernet0/1 are up, while the others are administratively down. The output of the 'sh ip route' command shows the routing table, including static routes for 50.0.0.0/8, 50.0.0.0/24, 50.0.1.0/25, 50.0.1.128/26, 50.0.1.192/30, and 50.0.1.196/30, all pointing to the next hop 50.0.1.194.

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#
Router#sh ip int br
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 50.0.1.129      YES manual up          up
FastEthernet0/1 50.0.1.193      YES manual up          up
Ethernet1/0      unassigned      YES unset   administratively down down
Ethernet1/1      unassigned      YES unset   administratively down down
Ethernet1/2      unassigned      YES unset   administratively down down
Ethernet1/3      unassigned      YES unset   administratively down down
Router#
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  50.0.0.0/8 is variably subnetted, 5 subnets, 4 masks
S       50.0.0.0/24 [1/0] via 50.0.1.194
S       50.0.1.0/25 [1/0] via 50.0.1.194
C       50.0.1.128/26 is directly connected, FastEthernet0/0
C       50.0.1.192/30 is directly connected, FastEthernet0/1
S       50.0.1.196/30 [1/0] via 50.0.1.194

Router#
Router#
Router#
Router#
```

Figure 8 — Router0: interface status (sh ip int br) and routing table (sh ip route) showing static routes to the Production and Sales networks.

Router0 also runs the DHCP server for the HR and Production pools. show ip dhcp pool confirms the HR pool covers 50.0.1.129–190 (62 usable addresses) and the Production pool covers 50.0.1.1–126 (126 usable addresses), matching the VLSM plan.

```

Router0
Physical Config CLI Attributes
IOS Command Line

Router#sh ip dhcp pool

Pool HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 62
Leased addresses : 2
Excluded addresses : 0
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
50.0.1.129 50.0.1.129 - 50.0.1.190 2 / 0 / 62

Pool Pro :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 126
Leased addresses : 2
Excluded addresses : 0
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
50.0.1.1 50.0.1.1 - 50.0.1.126 2 / 0 / 126

```

Figure 9 — DHCP pool statistics for HR and Production, each with 2 leased addresses so far (one PC and one laptop per department).

show ip dhcp binding lists every lease Router0 has handed out, tying each client's MAC address to the IP it was assigned — useful for confirming DHCP is working correctly across all three departments from a single router.

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 5 subnets, 4 masks
S 50.0.0.0/24 [1/0] via 50.0.1.194
S 50.0.1.0/25 [1/0] via 50.0.1.194
C 50.0.1.128/26 is directly connected, FastEthernet0/0
C 50.0.1.192/30 is directly connected, FastEthernet0/1
S 50.0.1.196/30 [1/0] via 50.0.1.194

Router#
Router#
Router#
Router#sh ip dhcp binding
IP address Client-ID/ Hardware address Lease expiration Type
50.0.1.130 00D0.D3D9.C010 -- Automatic
50.0.1.131 000C.85B6.7A15 -- Automatic
50.0.1.2 000C.CF95.1CB5 -- Automatic
50.0.1.3 0005.5EB7.6B15 -- Automatic
50.0.0.2 0060.7027.651C -- Automatic
50.0.0.3 0060.3E30.4080 -- Automatic
Router#

```

Figure 10 — Routing table repeated, followed by the full DHCP binding table: HR, Production, and Sales clients all present.

```

Router0
Physical Config CLI Attributes
IOS Command Li
Excluded addresses : 0
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
50.0.1.129 50.0.1.129 - 50.0.1.190 2 / 0 / 62

Pool Pro :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 126
Leased addresses : 2
Excluded addresses : 0
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
50.0.1.1 50.0.1.1 - 50.0.1.126 2 / 0 / 126

Pool Sales :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 2
Excluded addresses : 0
Pending event : none

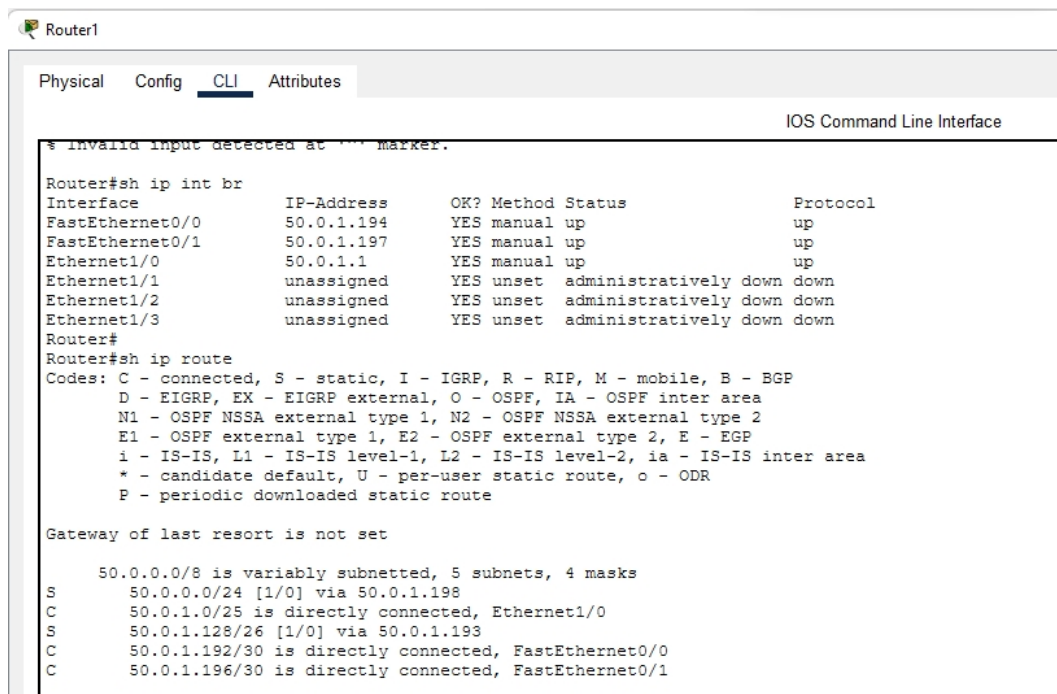
1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
50.0.0.1 50.0.0.1 - 50.0.0.254 2 / 0 / 254
Router#
Router#

```

Figure 11 — DHCP pool statistics continued, including the Sales pool (50.0.0.1–254, 254 usable addresses).

## 5.2 Router1 (Production / Middle Router)

Router1 sits in the middle of the chain. Its Ethernet1/0 interface (50.0.1.1) serves the Production LAN directly, while FastEthernet0/0 (50.0.1.194) links back to Router0 and FastEthernet0/1 (50.0.1.197) links onward to Router2. Static routes send HR-bound traffic out through 50.0.1.193 and Sales-bound traffic out through 50.0.1.198.



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

$ Invalid input detected at '^' marker.

Router#sh ip int br
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 50.0.1.194      YES manual up          up
FastEthernet0/1 50.0.1.197      YES manual up          up
Ethernet1/0     50.0.1.1        YES manual up          up
Ethernet1/1     unassigned      YES unset  administratively down down
Ethernet1/2     unassigned      YES unset  administratively down down
Ethernet1/3     unassigned      YES unset  administratively down down
Router#
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    50.0.0.0/8 is variably subnetted, 5 subnets, 4 masks
S       50.0.0.0/24 [1/0] via 50.0.1.198
C       50.0.1.0/25 is directly connected, Ethernet1/0
S       50.0.1.128/26 [1/0] via 50.0.1.193
C       50.0.1.192/30 is directly connected, FastEthernet0/0
C       50.0.1.196/30 is directly connected, FastEthernet0/1
```

Figure 12 — Router1: interfaces up on both WAN links and the Production LAN, with static routes to the HR (50.0.1.128/26) and Sales (50.0.0.0/24) networks.

## 5.3 Router2 (Sales Router)

Router2 serves the Sales LAN on FastEthernet0/1 (50.0.0.1) and connects back toward the rest of the network via FastEthernet0/0 (50.0.1.198). A single static default-style route sends all non-local traffic (HR, Production, and the HR–Production WAN link) toward 50.0.1.197 on Router1.

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

Router>
Router>enable
Router#sh ip int br
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 50.0.1.198      YES manual up          up
FastEthernet0/1 50.0.0.1        YES manual up          up
Router#
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      50.0.0.0/8 is variably subnetted, 5 subnets, 4 masks
C       50.0.0.0/24 is directly connected, FastEthernet0/1
S       50.0.1.0/25 [1/0] via 50.0.1.197
S       50.0.1.128/26 [1/0] via 50.0.1.197
S       50.0.1.192/30 [1/0] via 50.0.1.197
C       50.0.1.196/30 is directly connected, FastEthernet0/0

Router#
Router#
Router#
Router#
Router#
Router#
```

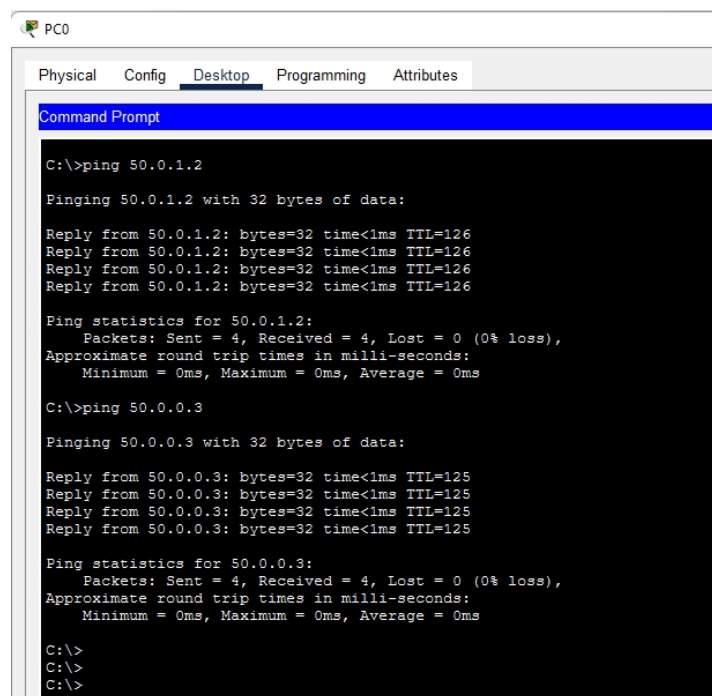
Figure 13 — Router2: interfaces up, with static routes back to the HR and Production subnets via Router1.

## 6. End-to-End Connectivity Testing (ping)

With addressing, DHCP, and static routing in place, the final step is to prove that devices in *different* departments — not just the same LAN — can actually reach each other. Each test below pings across at least one router hop, and every reply comes back with 0% packet loss, confirming the static routes on all three routers are correct and consistent.

### 6.1 PC0 (HR) → Production and Sales

From PC0 (50.0.1.130, HR), pings to PC2 (50.0.1.2, Production) and PC3 (50.0.0.3, Sales) both succeed with sub-millisecond round-trip times, showing traffic correctly crosses from the HR subnet, through Router0 and Router1, to reach both remote departments.

The image shows a screenshot of a PC0 desktop environment. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. Inside the window is a 'Command Prompt' application. The command prompt shows the following text:

```
C:\>ping 50.0.1.2

Pinging 50.0.1.2 with 32 bytes of data:

Reply from 50.0.1.2: bytes=32 time<1ms TTL=126
Reply from 50.0.1.2: bytes=32 time<1ms TTL=126
Reply from 50.0.1.2: bytes=32 time<1ms TTL=126
Reply from 50.0.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 50.0.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 50.0.0.3

Pinging 50.0.0.3 with 32 bytes of data:

Reply from 50.0.0.3: bytes=32 time<1ms TTL=125
Reply from 50.0.0.3: bytes=32 time<1ms TTL=125
Reply from 50.0.0.3: bytes=32 time<1ms TTL=125
Reply from 50.0.0.3: bytes=32 time<1ms TTL=125

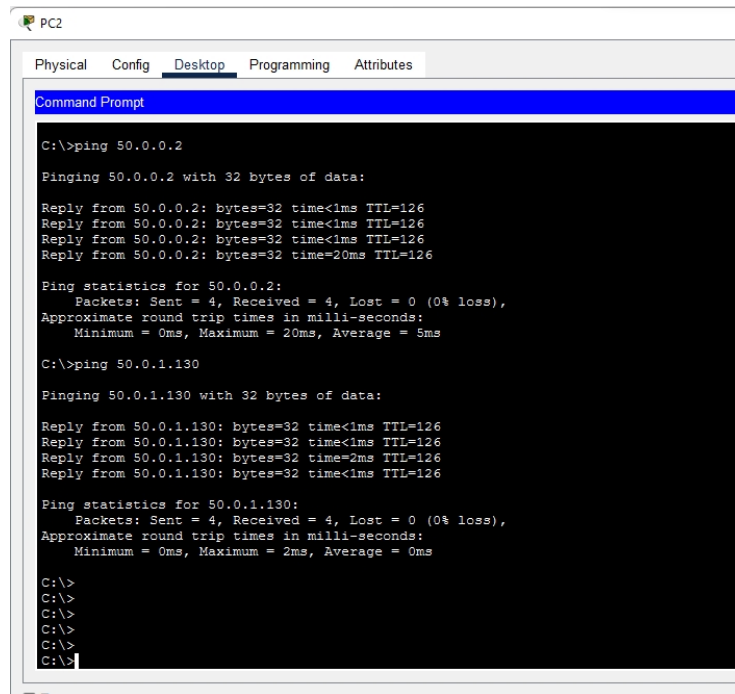
Ping statistics for 50.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>
C:\>
C:\>
```

Figure 14 — PC0 successfully pings 50.0.1.2 (Production) and 50.0.0.3 (Sales), 4/4 packets received on both, 0% loss.

### 6.2 PC2 (Production) → Sales and HR

From PC2 (50.0.1.2, Production), pings to 50.0.0.2 (Sales) and 50.0.1.130 (HR) both succeed, confirming Router1's static routes correctly forward traffic in both directions along the chain.



```
PC2
Physical Config Desktop Programming Attributes
Command Prompt

C:\>ping 50.0.0.2

Pinging 50.0.0.2 with 32 bytes of data:

Reply from 50.0.0.2: bytes=32 time<1ms TTL=126
Reply from 50.0.0.2: bytes=32 time<1ms TTL=126
Reply from 50.0.0.2: bytes=32 time<1ms TTL=126
Reply from 50.0.0.2: bytes=32 time=20ms TTL=126

Ping statistics for 50.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>ping 50.0.1.130

Pinging 50.0.1.130 with 32 bytes of data:

Reply from 50.0.1.130: bytes=32 time<1ms TTL=126
Reply from 50.0.1.130: bytes=32 time<1ms TTL=126
Reply from 50.0.1.130: bytes=32 time=2ms TTL=126
Reply from 50.0.1.130: bytes=32 time<1ms TTL=126

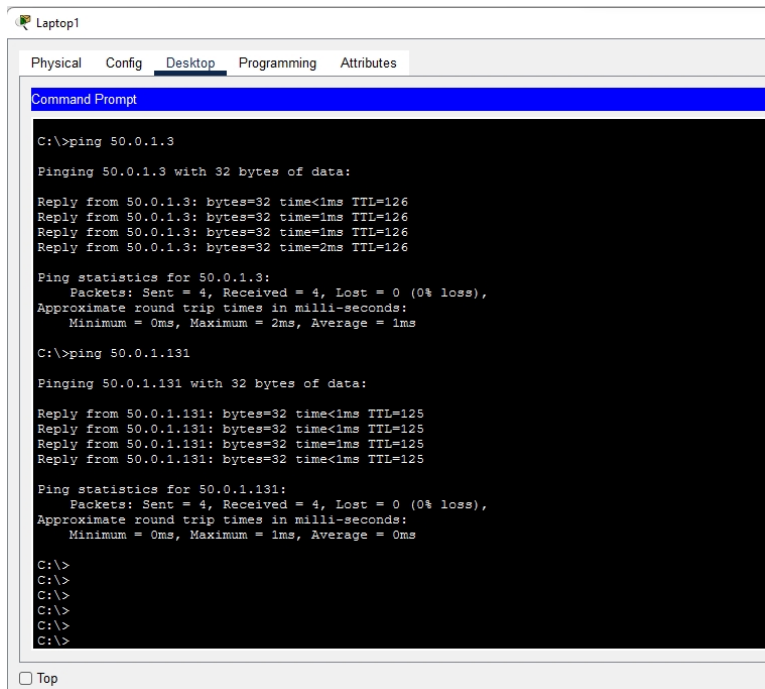
Ping statistics for 50.0.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
```

Figure 15 — PC2 successfully pings 50.0.0.2 (Sales) and 50.0.1.130 (HR), 4/4 packets received on both, 0% loss.

## 6.3 Laptop1 (Sales) → HR and Production

From Laptop1 (50.0.0.2, Sales), pings to 50.0.1.3 (Production) and 50.0.1.131 (HR) both succeed, completing the full mesh of cross-department connectivity tests: every department can now reach every other department.



```
Laptop1
Physical Config Desktop Programming Attributes
Command Prompt

C:\>ping 50.0.1.3

Pinging 50.0.1.3 with 32 bytes of data:

Reply from 50.0.1.3: bytes=32 time<1ms TTL=126
Reply from 50.0.1.3: bytes=32 time=1ms TTL=126
Reply from 50.0.1.3: bytes=32 time=1ms TTL=126
Reply from 50.0.1.3: bytes=32 time=2ms TTL=126

Ping statistics for 50.0.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>ping 50.0.1.131

Pinging 50.0.1.131 with 32 bytes of data:

Reply from 50.0.1.131: bytes=32 time<1ms TTL=125
Reply from 50.0.1.131: bytes=32 time<1ms TTL=125
Reply from 50.0.1.131: bytes=32 time=1ms TTL=125
Reply from 50.0.1.131: bytes=32 time<1ms TTL=125

Ping statistics for 50.0.1.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
```

Figure 16 — Laptop1 successfully pings 50.0.1.3 (Production) and 50.0.1.131 (HR), 4/4 packets received on both, 0% loss.

## 7. Conclusion

This project demonstrates the full workflow of building a routed, multi-department LAN in Packet Tracer: sizing subnets efficiently with VLSM, automating address assignment with per-department DHCP pools on a single router, linking three routers with static routes across two WAN point-to-point links, and validating the design end-to-end with ipconfig and ping. All six end devices obtained correct addresses, and every cross-department ping test passed with 0% packet loss, confirming the addressing plan and routing configuration are working as intended.